

Received 13 February 2026, accepted 28 March 2026, date of publication 1 April 2026, date of current version 6 April 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3679602

RESEARCH ARTICLE

Reaction-Uncertainty-Aware Motion Planning for Interactive Driving via Conditional Multi-Modal Motion Prediction

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This work was supported by the Ministry of Land, Infrastructure and Transport (MOLIT) of the Republic of Korea under Grant RS-2021-KA162184, and by the Ministry of Trade, Industry and Energy (MOTIE) and Korea Evaluation Institute of Industrial Technology (KEIT) of the Republic of Korea under Grant RS-2025-25448249.

ABSTRACT For autonomous vehicles to drive safely and in a socially collaborative manner, it is essential to consider interactions with surrounding agents. To model these interactions, recent motion planning works employ conditional motion predictions, which model the impact of the ego vehicle's plan on surrounding agents by treating the ego's future plan as query for prediction. However, most existing planning studies rely on uni-modal conditional motion prediction models, which struggle to represent the uncertain intentions of surrounding agents. Although a few studies adopt multi-modal mechanisms, they still fail to model reaction uncertainty—how surrounding agents respond to the ego's future plan. In this paper, we propose a reaction-uncertainty-aware motion planning framework that integrates a newly designed conditional multi-modal motion prediction (CMMP) with a tree-structured policy planner (TPP). CMMP effectively models the reaction uncertainty of surrounding agents through multi-modal and causal masked attention mechanisms. TPP generates tree structured ego-trajectory candidates and probabilistically evaluates them based on the multi-modal prediction results to enhance the robustness of decision-making. The proposed framework is validated on the large-scale real-world nuPlan dataset. Experimental results demonstrate that, compared to the baseline relying on uni-modal conditional motion prediction, our method achieves up to a 32.39% improvement in planning performance, particularly excelling in interactive scenarios.

INDEX TERMS Autonomous driving, motion planning, conditional motion prediction, integration of prediction and planning.

I. INTRODUCTION

Autonomous vehicles must operate safely and efficiently in dynamic environments shared with multiple traffic agents. Achieving this goal hinges on the vehicle's ability to reason about mutual influence between the ego vehicle and surrounding agents. This requires understanding how ego plans affect the behaviors of others and planning accordingly in anticipation of their reactions. However, traditional sequential pipelines [1], [2], [3], which modularize and sequentially connect prediction and planning, limit such reasoning due to their unidirectional information flow [4].

The associate editor coordinating the review of this manuscript and approving it for publication was Valentin Ivanov.

As an alternative, undirected architectures [5], [6], [7], [8], such as monolithic end-to-end (E2E) frameworks, have emerged. These approaches can implicitly encode interactions and uncertainty, but often suffer from low interpretability and limited reliability in safety-critical scenarios due to their black-box characteristics. Serving as a middle ground, bidirectional architectures [9], [10], [11], [12], [13] couple prediction and planning while retaining interpretability and reliability. These advantages are achieved through modularization and mutual information exchange between the prediction and planning, enabling the system to reflect interactions and adjust plans accordingly.

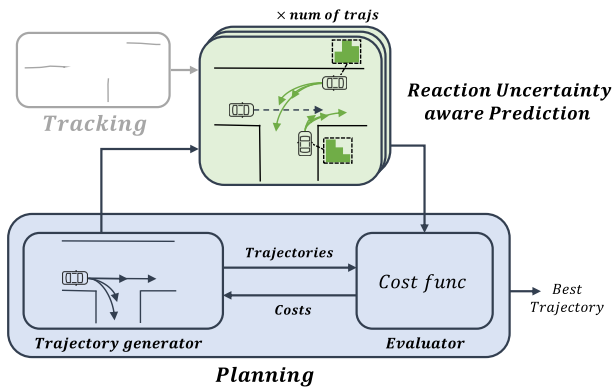


FIGURE 1. Overview of the proposed reaction-uncertainty-aware motion planning framework.

Within bidirectional architectures, it is essential to model the interactions between the ego vehicle and surrounding agents. Early approaches adopt rule-based methods [14], [15] such as the intelligent driver model (IDM) [16] to model how ego plans influence surrounding agents by using simple mathematical formulations. Whereas these methods are computationally efficient, they oversimplify interactions, limiting realism and generalization. Among them, conditional motion prediction (CMP) [17], [18], [19], [20], [21], [22], [23] models the impact of ego plans by treating predefined ego trajectories as queries for prediction. However, existing planning studies adopting CMP [11], [12], [13], [24] often rely on uni-modal predictions, overlooking the uncertain behaviors of surrounding agents [9]. Although a few planning works have adopted multi-modal mechanisms [24], they still fail to capture reaction uncertainty—how surrounding agents respond to the ego’s future plan, which is critical in interactive scenarios. These challenges highlight the need for a more reaction-aware framework to better support interactive autonomous driving.

In this paper, we propose a reaction-uncertainty-aware motion planning (RaMP) framework, composed of prediction and planning modules in a bidirectional structure, as illustrated in Fig. 1. Previous works, which have bidirectional frameworks, either rely on uni-modal predictions [11], [13] or fail to capture reaction uncertainty [24]. In contrast, our approach explicitly models reaction uncertainty through a newly designed prediction module, conditional multi-modal prediction (CMMP), which crucially employs a multi-modal cross-attention mechanism within the ego-conditioning network to capture the responses. This module is further integrated with a planner module, tree-structured policy planner (TPP), which generates ego-trajectory candidates and probabilistically evaluates them based on the multi-modal prediction results. The integration of CMMP and TPP enables robust, reaction-aware planning by effectively reasoning about how surrounding agents respond to the ego’s future plans and planning trajectories accordingly. The key contributions of this paper can be summarized below:

- A bidirectional motion planning framework is proposed, which integrates CMMP with TPP to effectively model the interactions between the ego vehicle and surrounding agents, supporting robust and reaction-aware planning.
- A novel transformer-based CMMP is designed to effectively capture the reaction uncertainty of surrounding agents while efficiently modeling the diverse outcomes induced by different ego plans.
- The proposed method outperforms baselines in closed-loop evaluations on the real-world dataset, particularly in highly interactive scenarios, demonstrating its effectiveness in handling agent interactions.

II. RELATED WORK

A. INTERACTION-AWARE MOTION PREDICTION

Deep learning-based motion prediction has rapidly advanced with the availability of large-scale driving datasets [25], [26]. Many approaches leverage architectures such as graph neural networks (GNNs) [27], [28] and transformers [29], [30] to model interactions with road structures and multiple agents. In particular, interactive scenarios require modeling how the ego vehicle’s plan affects surrounding agents. To account for the influence, conditional motion prediction (CMP) has emerged as a promising approach, incorporating the ego vehicle’s future trajectory as a conditioning signal to improve prediction accuracy [17], [18], [19], [20], [21], [22], [23]. whereas this enhances accuracy, most CMP methods rely on uni-modal prediction schemes, limiting their ability to capture the stochastic behaviors of surrounding agents. Recent efforts [21], [22], [23] introduce conditional multi-modal motion prediction (CMMP) to capture the uncertain intentions of surrounding agents. However, these methods primarily focus on uncertainty arising from agent-map or agent-agent interactions based on historical states, using attention mechanisms or convolutional approaches. As a result, they often fail to account for reaction uncertainty—how surrounding agents respond to the ego’s future plan. Furthermore, aside from [24], they have never been integrated with downstream planning modules nor have they considered diverse ego-trajectory candidates. Even that study [24] lacks validation in closed-loop settings, leaving its effectiveness in real-world planning applications unverified. In contrast, our proposed CMMP explicitly models reaction uncertainty using a multi-modal cross-attention mechanism within the ego-conditioning network. Moreover, it considers diverse future scenarios from different ego trajectories, is tightly integrated with a downstream planner, and is validated in closed-loop settings, demonstrating its effectiveness in real-world planning applications.

B. INTERACTION-AWARE MOTION PLANNING

Autonomous driving in interactive scenarios requires reasoning about how the ego vehicle’s actions influence surrounding agents and planning accordingly. Early end-to-end behavior cloning approaches [6], [7], [8] learn implicit interactions

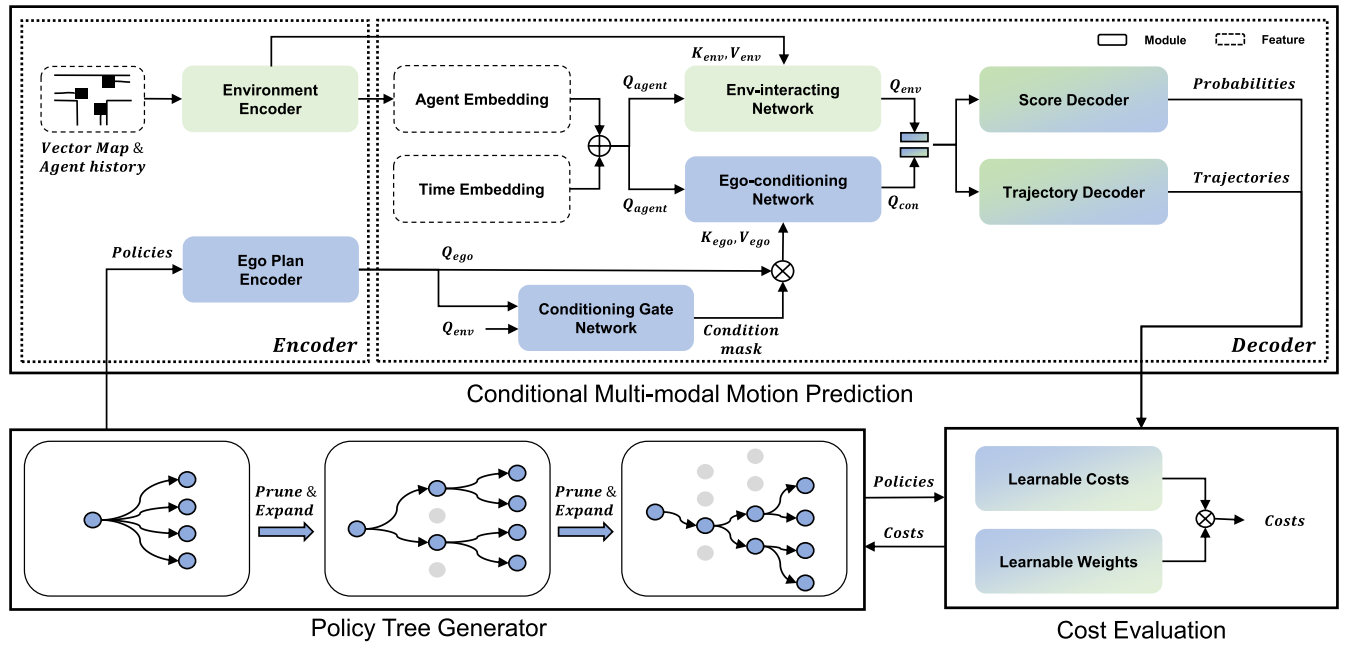


FIGURE 2. RaMP framework consists of a conditional multi-modal motion prediction (CMMP) module and a tree policy planner (TPP), which is composed of a policy tree generator and a cost evaluation module. The prediction module takes trajectory candidates generated by the tree policy planner as conditional inputs and predicts multi-modal reaction trajectories of surrounding agents with associated probabilities. The cost evaluation module evaluates each trajectory candidate using the predicted reaction outcomes and selects the best trajectory based on the expected cost.

using a single neural network but lack interpretability and modular control. Joint optimization methods [10], [31], [32], [33] offer a more interpretable structure by explicitly modeling multi-agent futures, but they assume uni-modal behaviors that often fail in uncertain real-world traffic situations. In contrast, tree-structured policy planners (TPPs) [11], [14], [15] represent future scenarios as trees conditioned on each ego's plans, enabling more interpretable, modular control and reactive planning. However, most studies suffer from iterative computations for conditioning multiple ego plans, leading to high computational overhead. Although Huang et al. [13] reduce this by utilizing a causal masked cross-attention for efficient ego conditioning, their method adopts uni-modal prediction, making it vulnerable to the uncertain intentions of surrounding agents. Huang et al. [24] also introduce CMMP and integrate it with a sampling-based planner, but it still requires iterative calculations for conditioning multiple ego plans, and fails to model reaction uncertainty.

In parallel, another challenge of TPPs is their dependence on hand-crafted cost functions, which limit adaptability. Through inverse reinforcement learning, some works [13], [24] attempt to learn such costs from data, improving generalization to complex scenarios. However, no existing framework simultaneously captures the reaction uncertainty of surrounding agents and accounts for stochastic behaviors in the cost evaluation. To this end, we propose a reaction-uncertainty aware motion planning framework that combines a newly designed CMMP with a TPP, enabling efficient multiple ego-conditioning, reaction uncertainty modeling, and robust decision-making.

III. METHODOLOGY

The detailed architecture of the proposed RaMP framework is illustrated in Fig. 2. It consists of three main components: a policy tree generator that samples multiple ego-trajectory candidates; a conditional multi-modal motion prediction (CMMP) that predicts surrounding agent future behaviors conditioned on each ego-trajectory; and a cost evaluation that selects the final plan by evaluating the candidates based on motion features and prediction results. Our method builds upon DTPP [13], introducing key modifications to prediction and cost evaluation. Specifically, we design a novel CMMP to capture the reaction uncertainty of surrounding agents while preserving efficient ego-conditioning on multiple plans. Furthermore, we extend the cost function to incorporate multi-modal predictions, enabling more stable and robust decision-making in interactive scenarios.

A. PROBLEM FORMULATION

Consider an autonomous driving scenario with one ego vehicle and N_n surrounding agents. Let S^e and $S^{\text{sur}} = \{S_1^{\text{sur}}, \dots, S_{N_n}^{\text{sur}}\}$ denote the current states of the ego vehicle and surrounding agents, respectively. In addition, let S^{hist} represent the historical states of all agents up to the current timestep, and let M denote a high-definition map. The goal of RaMP is to plan a future trajectory for the ego vehicle that accounts for multi-modal prediction results of surrounding agents. To this end, a policy tree model generates a set of N_b candidate future trajectories for the ego vehicle, denoted as $\zeta^e = \{\zeta_1^e, \dots, \zeta_{N_b}^e\} = \text{PolicyTree}(S^e, M)$. For each ego candidate ζ_i^e , the conditional multi-modal

motion prediction (CMMP) predicts the future behaviors of each surrounding agent j by generating N_m multi-modal trajectories $\zeta_{i,j}^{\text{sur}} = \{\zeta_{i,j,1}^{\text{sur}}, \dots, \zeta_{i,j,N_m}^{\text{sur}}\}$ along with their probabilities $p(\zeta_{i,j}^{\text{sur}}) = \{p(\zeta_{i,j,1}^{\text{sur}}), \dots, p(\zeta_{i,j,N_m}^{\text{sur}})\}$, where all predictions are conditioned on the ego-trajectory ζ_i^e and the map M , i.e., $(\zeta_{i,j}^{\text{sur}}, p(\zeta_{i,j}^{\text{sur}})) = \text{CMMP}(\zeta_j^{\text{sur}} | S^{\text{hist}}, M, \zeta_i^e)$. Each ego-trajectory candidate ζ_i^e is then evaluated using a cost function $c_i^e = \text{CostEvaluation}(\zeta_i^e, \{\zeta_{i,j}^{\text{sur}}\}_{j=1}^{N_n}, M)$, which reflects planning objectives such as safety, efficiency, and comfort under the predicted interactions. The optimal trajectory is finally selected by minimizing the cost across all candidates, i.e., $\zeta^{e*} = \arg \min_{\zeta_i^e \in \zeta^e} c_i^e$.

B. POLICY TREE GENERATOR

The policy tree generator produces trajectory candidates for the ego vehicle using a sampling-based method. These candidates are generated along the centerline of each lane, enabling the ego vehicle to plan a trajectory that follows the lanes. The policy tree planning process consists of path generation and speed planning. In the path-generation part, terminal states are determined by sampling points along the centerlines. Then, continuous paths connecting the initial and terminal states are generated using a cubic Bézier curve. Once the paths are generated, terminal velocities are sampled, and an equivalent acceleration is applied from the initial velocity to generate trajectory candidates. These candidates form a tree structure with short-term (3s) and long-term (5s) stages. The generator first produces short-term trajectory candidates, which are scored and pruned by the cost evaluator. The remaining candidates are then expanded into long-term trajectories for final selection.

C. CONDITIONAL MULTI-MODAL MOTION PREDICTION

1) ENCODER

The encoder module extracts meaningful low-level features from the historical states of the ego vehicle, surrounding agents, and vectorized map. It comprises two parts: the environment encoder and the ego plan encoder. The environment encoder uses LSTM networks to process agent trajectories and encodes map elements (e.g., lanes, crosswalks, traffic signals) via MLPs, followed by max-pooling for spatial aggregation. To capture inter-dependencies among multiple agents and the map, it processes the environment context using a self-attention layer [34]. Meanwhile, the ego plan encoder encodes ego-trajectory candidates from the policy tree generator using MLPs to produce low-level features. Outputs from both encoders serve as the query, key, and value for the following decoder layer.

2) DECODER

The decoder takes low-level features from the encoder and generates future trajectories for surrounding agents as a set of multiple trajectories with associated probabilities. It consists of three core components—an environment-interacting

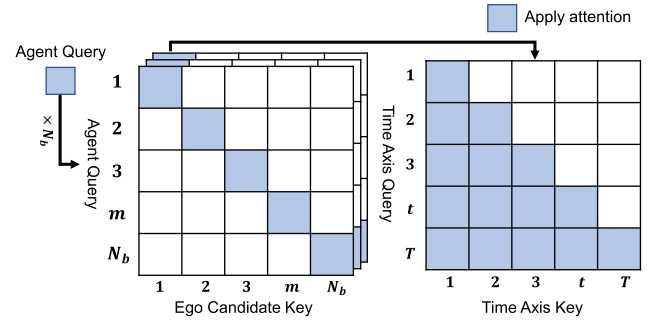


FIGURE 3. Multi-modal and causal masked attention within the ego-conditioning network. Multi-modal attention represents reaction uncertainty conditioned on a given ego-trajectory, while causal masked attention captures temporal dependencies.

network, an ego-conditioning network, and a conditioning gate network—followed by two decoding heads—a trajectory decoder and a score decoder.

a: INTERACTION MODELING

whereas the encoder captures the environment context via self-attention, the decoder further refines these relations through two cross-attention-based interaction modules. The env-interacting network encodes contextual interactions among agents and map elements, using the encoder output as key-value pairs ($K_{\text{env}}, V_{\text{env}}$). The ego-conditioning network models how each ego-trajectory candidate influences surrounding agents by attending to the ego-conditioned features ($K_{\text{ego}}, V_{\text{ego}}$). Both modules use an agent motion query Q_{agent} that combines agent embeddings with time embeddings.

b: REACTION UNCERTAINTY MODELING

Unlike prior works [24], [30], [35] that focus on agent-agent or agent-map uncertainty, our framework targets reaction uncertainty. To model the reaction uncertainty—how surrounding agents respond to the ego vehicle’s behavior—while maintaining computational efficiency when conditioning on multiple ego plans, we incorporate multi-modal attention [30] and causal masked attention [13] into the ego-conditioning network, as illustrated in Fig. 3. The network comprises N_m mode-specific branches, each modeling a distinct ego-agent interaction pattern. In our experiments, we set $N_m = 3$. To ensure that the model learns diverse reactions, the loss is propagated only through the specific mode network branch whose prediction is closest to the ground truth. To condition on multiple ego plans in parallel, each agent’s query is broadcast across all ego-trajectory candidates, N_b , and causal masked cross-attention is applied. Moreover, to prevent irrelevant ego conditioning, we introduce an MLP-based conditioning gate network that predicts a condition mask to modulate Q_{ego} before being fused with Q_{agent} in the ego-conditioning network. This mechanism ensures that the model attends to ego-agent interactions only when semantically meaningful.

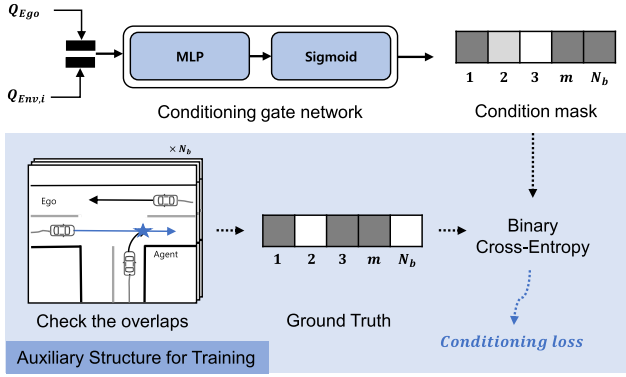


FIGURE 4. Structure for training the condition mask. The mask is supervised using ego-agent overlap-based ground truth with a binary cross-entropy loss. This enables the model to predict ego-agent relevance and adaptively control the degree of ego conditioning.

c: EGO CONDITION MASKING

To prevent irrelevant ego conditioning that may hinder feature disentanglement, we introduce an MLP-based conditioning gate network, as illustrated in Fig. 4. This network dynamically learns to predict a condition mask, G^{cond} , that selectively modulates the ego-specific query feature Q_{ego} before it is fused with the agent-related query Q_{agent} . By applying this gating operation, the model can adaptively determine which agents and to what extent the ego information should influence within the overall interaction representation. In essence, the gate suppresses redundant or noisy ego cues while emphasizing informative ones, ensuring that the network focuses on semantically meaningful ego-agent interactions. As a result, the proposed mechanism enhances both generalization across diverse driving scenarios and robustness under varying traffic conditions.

d: TRAJECTORY AND SCORE DECODING

The output features from the interaction modules, Q_{env} and Q_{con} , are fused and passed to two decoding heads. The trajectory decoder, implemented as an MLP, produces multiple future trajectories for each agent. Since the ego-conditioning network produces modality-specific features, the output from the env-interacting network, Q_{env} , is broadcast to match them. The score decoder compresses temporal information via convolution, followed by MLPs and a softmax layer to generate the probability for each predicted trajectory.

D. COST EVALUATION

The cost evaluation module assesses ego-trajectory candidates and selects the final ego-trajectory from them. We use an IRL-based approach that employs learnable features and weights. Compared to the baseline, DTPP [13], which evaluates each trajectory using uni-modal prediction results and thus ignores uncertain intentions of surrounding agents, our method incorporates multi-modal predicted trajectories with associated probabilities during cost evaluation. This

probabilistic evaluation accounts for the agents' uncertain intentions and enhances planning robustness and stability.

1) COST FUNCTION

The cost function is defined as a weighted sum of feature terms over the planning horizon, as shown in (1). It consists of three features: Ego feature, f^E , representing ego-related factors such as speed and comfort; Interaction feature, f^I , capturing interaction dynamics with surrounding agents; Collision feature, f^C , assessing potential collision risks. Unlike the baseline, our method incorporates multi-modal trajectories with their associated probabilities into the interaction feature f^I , and the collision feature f^C .

$$c = \frac{1}{t_F - t_0} \left(w_E f^E + w_I f^I + w_C f^C \right). \quad (1)$$

2) EGO FEATURE

The ego feature is designed to evaluate the comfort and efficiency of the trajectory. It includes the square of speed, acceleration, and jerk along the planned trajectory. These features are designed to capture the dynamics of the ego vehicle while balancing both driving comfort and progress capability. By incorporating these metrics, the model ensures a comprehensive evaluation of the ego vehicle's motion characteristics, enabling smoother and more efficient trajectory planning.

3) INTERACTION FEATURE

The interaction feature is designed to model the interaction dynamics between the ego vehicle and surrounding agents, as shown in (2). It uses relative states between the ego plan and the predicted multi-modal trajectories of surrounding agents. These relative states are encoded through an MLP and weighted by each predicted trajectory's probability, p . The resulting features are aggregated via max pooling and passed through another MLP to obtain a compact interaction representation. By incorporating multi-modal predictions, the feature is able to more accurately model interaction dynamics.

$$f^I = \sum_{j=1}^{N_n} \sum_{k=1}^{N_m} \text{MLP} \left(\text{MaxPool} \left(p(\zeta_{i,j,k}^{sur}) \cdot \text{MLP}(\zeta_i^e - \zeta_{i,j,k}^{sur}) \right) \right), \quad (2)$$

where N_n denotes the number of surrounding agents, N_m the number of modality, ζ_i^e an ego-trajectory candidate, and $\zeta_{i,j,k}^{sur}$ the predicted trajectory conditioned on ζ_i^e .

4) COLLISION FEATURE

The collision feature is designed to evaluate collision risk, as shown in (3). It is calculated based on the distance between the ego vehicle and surrounding agents over the planning horizon. An exponential function is applied to the distance between the ego and each surrounding agent to produce a collision score, which is then weighted by its predicted

probability, p , to account for uncertainty in agents' future behaviors.

$$f^C = \sum_{j=1}^{N_n} \sum_{k=1}^{N_m} p(\xi_{i,j,k}^{sur}) \cdot \exp\left(-0.2 \cdot \|\xi_i^e - \xi_{i,j,k}^{sur}\|^2\right). \quad (3)$$

E. OBJECTIVE FUNCTION FOR TRAINING

The conditional multi-modal motion prediction and cost evaluation are fully differentiable, allowing the entire framework to be trained end-to-end using gradient-based optimization. The training objective consists of four components: a prediction loss, L_p ; an IRL loss, L_I ; a modality loss, L_m ; and a condition mask loss, L_c . The total loss is formulated as:

$$L = \sum_{s=1}^{N_s} \delta^s (L_p + \alpha L_I + \beta L_m + \gamma L_c), \quad (4)$$

where N_s denotes the number of planning stages and δ^s is a stage-specific weight that adjusts each stage's contribution to the total loss. In our experiments, δ^1 is set to 1.0 and δ^2 to 0.2. The hyperparameters α , β , and γ are set to 1.0, 2.0, and 10.0, respectively.

1) PREDICTION LOSS

The prediction loss is designed to improve the trajectory prediction accuracy of surrounding agents. It is defined using the smooth L1 loss between the predicted trajectory of each surrounding agent and its ground truth future trajectory:

$$L_p = \frac{1}{t_F - t_0} \cdot \frac{1}{N_n} \sum_{j=1}^{N_n} \text{Smooth L1}(\xi_j^{sur*} - \xi_j^*), \quad (5)$$

where ξ_j^* denotes the ground truth trajectory of the agent, ξ_j^{sur*} represents the most probable predicted trajectory conditioned on ξ^{e*} , which corresponds to the ego candidate closest to the expert demonstration.

2) IRL LOSS

The IRL loss is designed to guide the autonomous vehicle toward expert-like behavior by learning human driving costs from the demonstration. The loss is computed by first converting the costs of ego-trajectory candidates into a probability distribution using a softmax over negative costs:

$$P(\xi_i^e) = \frac{e^{-c(\xi_i^e)}}{\sum_{i=1}^{N_b} e^{-c(\xi_i^e)}}, \quad (6)$$

where, N_b denotes the number of ego-trajectory candidates, ξ_i^e denotes ego-trajectory candidate. A cross-entropy loss is then applied with respect to the ground truth label, y_i^{irl} , which is a one-hot vector indicating the candidate closest to the expert demonstration:

$$L_I = - \sum_{i=1}^{N_b} y_i^{irl} \log(p(\xi_i^e)). \quad (7)$$

3) MODALITY LOSS

The modality loss is designed to train the probability distribution over each predicted trajectory for surrounding agents. It is calculated as a binary cross-entropy loss between the predicted probability, $p(\xi_{j,k}^{sur})$ of each trajectory and the ground truth label, $y_{j,k}^m$, which is a one-hot vector representing the trajectory that is closest to the actual future trajectory:

$$L_m = - \sum_{j=1}^{N_n} \sum_{k=1}^{N_m} y_{j,k}^m \log(p(\xi_{j,k}^{sur})). \quad (8)$$

4) CONDITION MASK LOSS

The condition mask loss is designed to train the conditioning gate network. It is computed using binary cross-entropy between the predicted condition mask, $G_{i,j}^{cond}$ and the ground truth label, $y_{i,j}^{gate}$, which is calculated based on whether the trajectories of the ego candidate and a surrounding agent overlap within a predefined distance during the planning horizon:

$$L_c = - \sum_{i=1}^{N_b} \sum_{j=1}^{N_n} y_{i,j}^{gate} \log(p(G_{i,j}^{cond})). \quad (9)$$

IV. EXPERIMENT AND RESULTS

A. EXPERIMENTAL SETUP

The proposed method is trained and evaluated on the large-scale real-world nuPlan dataset [25]. A total of 500,000 scenarios from the training split are used for training, whereas the test split is used for evaluation. Experiments are conducted for both prediction and planning tasks. Prediction performance is evaluated in an open-loop setting using the average displacement error (ADE) and final displacement error (FDE) over an 8-second prediction horizon. For planning, we follow nuPlan's closed-loop evaluation protocol [25], which aggregates safety, efficiency, and comfort metrics. Notably, safety is applied as a multiplicative factor, thereby significantly impacting the overall score. Detailed descriptions of the evaluation metrics are provided in Appendix A. To assess performance under varying levels of interaction complexity, we conduct experiments on three nuPlan benchmarks: **test14-random**, which contains standard driving scenarios; **test14-hard** [39], which focuses on challenging cases; and **test14-inter** [40], which consists of highly interactive scenarios where modeling agent reactions is critical. In addition, two closed-loop simulation settings are considered: CL-NR (closed-loop non-reactive), in which surrounding agents follow log data, and CL-R (closed-loop reactive), in which surrounding vehicles respond to the ego vehicle using the Intelligent Driver Model (IDM) [16].

B. IMPLEMENTATION DETAILS

For planning, we adopt a two-stage hierarchical strategy over an 8-second horizon. The planning process first operates over a short-term interval of 0 to 3 seconds, during which three candidate reference paths are considered and up to 30 target

TABLE 1. Closed-loop planning performance in nuPlan simulation.

Method	Test14-random			Test14-hard			Test14-inter		
	CL-NR $\uparrow$	CL-R $\uparrow$	Avg. $\uparrow$	CL-NR $\uparrow$	CL-R $\uparrow$	Avg. $\uparrow$	CL-NR $\uparrow$	CL-R	Avg. $\uparrow$
RasterModel [25]	69.66	67.54	68.6	49.47	52.16	50.82	57.62	59.46	58.54
UrbanDriver [36]	63.27	61.02	62.14	51.54	49.07	50.31	63.89	59.98	61.94
GC-PGP [37]	55.99	51.39	53.69	43.22	39.63	41.43	47.82	49.66	48.74
PDM-Open [38]	52.80	57.23	55.02	33.51	35.83	34.67	33.56	42.79	38.18
PlanTF [39]	87.07	86.48	86.78	72.68	61.7	67.19	<u>78.33</u>	76.53	77.43
DTPP [13]	70.32	76.29	73.31	58.05	63.31	60.68	59.17	70.62	64.90
RaMP w/o gate	80.88	77.43	79.16	<u>66.79</u>	<u>64.35</u>	<u>65.57</u>	78.34	82.81	80.58
RaMP	80.53	<u>79.51</u>	<u>80.02</u>	64.39	64.61	64.5	75.83	<u>80.37</u>	<u>78.1</u>

CL-NR: closed-loop non-reactive simulation, CL-R: closed-loop reactive simulation.

states are sampled. Based on the resulting predictions, only the top five candidate nodes are selected. Each of these nodes is then further expanded in a second, long-term stage spanning 3 to 8 seconds, during which six target states are generated per node, resulting in a maximum of 30 trajectory branches overall. The model is trained on a single NVIDIA RTX 3090 GPU with a batch size of 16 for 20 epochs. We use the AdamW optimizer with an initial learning rate of 1×10^{-4} , which is halved every five epochs to promote stable convergence. Inference is performed under the same configuration.

C. COMPARED METHODS

We compare our method against a range of planners, including raster-, vector-, graph-, and transformer-based approaches. RasterModel [25] is a CNN-based planner from nuPlan that predicts future waypoints directly from rasterized BEV features without explicit ego state input. UrbanDriver [36] uses a PointNet-style encoder with attention to capture agent-map interactions. GC-PGP [37] generates trajectory proposals constrained by lane graphs and selects the most likely cluster center. PDM-Open [38] is an MLP-based model using vectorized agent and map features. PlanTF [39] combines a transformer encoder with augmented training data and an additional reinforcement learning adapter. DTPP [13], our baseline, uses uni-modal conditional motion prediction (CMP) and utilizes a tree-structured policy planner (TPP).

D. MAIN RESULTS

1) CLOSED-LOOP PLANNING

Table 1 presents closed-loop planning results on the Test14-random, Test14-hard, and Test14-inter benchmarks in the nuPlan simulation. Our method, RaMP, consistently outperforms learning-based methods that rely solely on behavior cloning [25], [36], [37], [38], which often suffer from covariate shifts in unseen environments. Our method achieves outstanding performance in interactive scenarios, achieving the best results on the Test14-inter benchmark, where agent interactions are crucial. Since safety metrics are multiplicatively aggregated in the closed-loop score, the strong

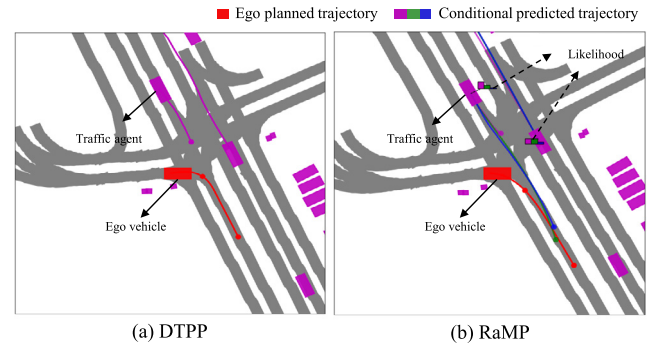


FIGURE 5. Comparison with DTPP. DTPP predicts a single deterministic trajectory for each agent, whereas RaMP predicts multiple plausible trajectories with their probabilities by explicitly modeling reaction uncertainty.

Test14-inter performance highlights RaMP's effectiveness in preserving safety during complex agent interactions. This result indicates that RaMP generates safer, more interaction-aware trajectories, effectively reducing collision risk in densely interactive scenarios. Specifically, RaMP outperforms planTF [39], the current top-performing method on the leaderboard. Moreover, unlike other high-performing approaches such as planTF and UrbanDriver that degrade when switching from CL-NR to CL-R simulation, RaMP further improves, demonstrating its superior ability to model interactions with surrounding agents through conditional motion prediction and its bidirectional architecture. Whereas RaMP excels in interactive scenarios, planTF performs better in less interactive scenarios, such as Test14-random and Test14-hard. However, planTF's performance relies on augmented training data and an additional reinforcement learning-based controller. In contrast, RaMP achieves competitive results using raw data and a standard LQR controller. Additionally, RaMP consistently outperforms our baseline, DTPP [13], as further discussed in the next section.

2) COMPARISON WITH BASELINE

Fig. 5a and Fig. 5b present qualitative comparisons with the baseline DTPP [13], which employs a uni-modal conditional prediction module that predicts a single trajectory per agent

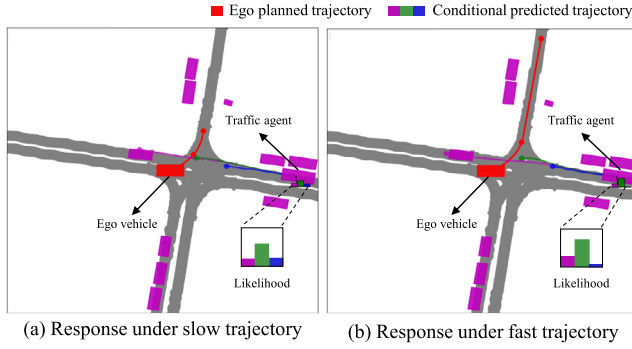


FIGURE 6. Conditional prediction results under different ego plans. RaMP produces distinct reaction distributions depending on the ego-trajectory. For a slower ego-trajectory, higher probabilities are assigned to conservative agent motions (green and blue), resulting in safer interactions; for a faster ego-trajectory, higher probabilities are assigned to more aggressive agent motions (purple and fast green), leading to more efficient interactions.

for each ego plan. Although DTPP leverages multiple outcomes conditioned on different ego plans during cost evaluation, each outcome is deterministically predicted, thereby failing to capture the multi-modal nature of the agent's reactions to a specific ego plan. In contrast, our method models uncertain reactions using a multi-modal cross-attention mechanism within the ego-conditioning network and produces multiple trajectories with associated probabilities. By integrating these probabilistic forecasts into the planner's cost evaluation, our approach accounts for a range of potential agent behaviors, leading to more robust, informed decision-making. As shown in Table 1, the proposed method consistently outperforms DTPP across all benchmarks, achieving up to a 32.39% improvement—especially in interactive scenarios. These results validate the effectiveness of modeling reaction uncertainty and probabilistically integrating it into the planning pipeline, enabling more robust and interactive motion planning. For further qualitative comparisons on the nuPlan simulation, please refer to Appendix.

3) EGO-CONDITIONED PREDICTION

Fig. 6a and Fig. 6b illustrate conditional prediction outputs under different ego plans in an intersection scenario. Fig. 6a corresponds to a slower ego-trajectory candidate, whereas Fig. 6b shows results for a faster one. The proposed method predicts the behavior of a vehicle approaching from the opposite direction that interacts directly with the ego vehicle. Under the slower ego plan (Fig. 6a), the model assigns higher probabilities to safer responses (blue and green) and lower to the riskier trajectory (purple). Conversely, under the faster one (Fig. 6b), the predicted distribution shifts toward faster, more efficient responses (purple, green). These results demonstrate the model's ability to produce reasonable ego-conditioned multi-modal predictions, allowing the planner to consider diverse outcomes from surrounding agents under varying ego plans.

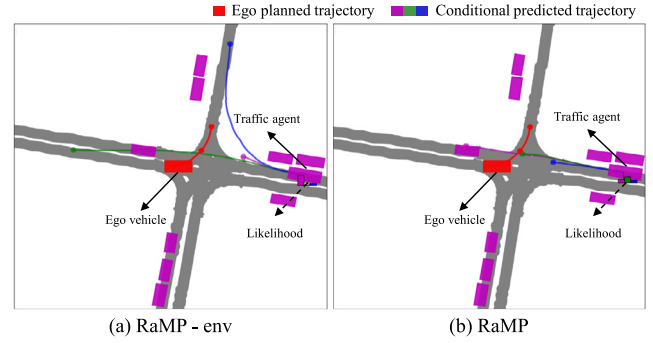


FIGURE 7. Qualitative comparison of uncertainty-modeling architectures. RaMP-env captures map- and agent-related uncertainties using a multi-modal attention mechanism within the environment-interacting network, whereas RaMP focuses on modeling reaction uncertainty of surrounding agents, which is critical for planning in highly interactive scenarios.

TABLE 2. Quantitative comparison of uncertainty-modeling architectures.

Method	Test14-random		Test14-hard		Test14-inter	
	CL-NR ↑	CL-R ↑	CL-NR ↑	CL-R ↑	CL-NR ↑	CL-R
RaMP-env	74.71	74.42	60.27	56.29	71.45	76.26
RaMP	80.53	79.51	64.39	64.61	75.83	80.37

E. ABLATION STUDIES

1) EFFECTIVENESS OF MODELING REACTION UNCERTAINTY
Prior works on multi-modal motion prediction [21], [24], [30] primarily capture uncertainty from agent-agent or agent-map interactions, leveraging historical states and map features. In contrast, our method models reaction-level uncertainty by incorporating multi-modal cross-attention within the ego-conditioning network to model how surrounding agents respond to the ego's future plan. To evaluate the effect of different uncertainty sources, we introduce a variant—RaMP-env—which applies the same multi-modal attention mechanism to an env (agent & map)-interacting network and exclude the conditioning gate network, while keeping all other components identical. Fig. 7 qualitatively compares the two architectures. RaMP predicts diverse reaction behaviors, such as yielding or hesitation, whereas RaMP-env tends to predict geometric intentions, such as turning or going straight. These differences are reflected in the closed-loop planning results in Table 2, where RaMP consistently outperforms RaMP-env. These findings highlight that modeling reaction-level uncertainty is more crucial for motion planning than focusing on agent-agent or agent-map interaction uncertainty.

2) EFFECTIVENESS OF NON-ITERATIVE EGO-CONDITIONING
Prior works on conditional motion prediction [17], [18], [19], [20], [21], [22], [23] and using it with a planner [11], [12], [24] adopt an iterative ego-conditioning framework, in which the model repeatedly predicts agent behavior conditioned on diverse ego plans. Although these methods are

TABLE 3. Runtime analysis (s) of prediction models.

Method	Encode	Expand	Predict	Expand	Predict	Total
RaMP-iter	0.007	0.013	0.705	0.015	0.750	1.490
RaMP	0.007	0.013	0.141	0.015	0.103	0.278

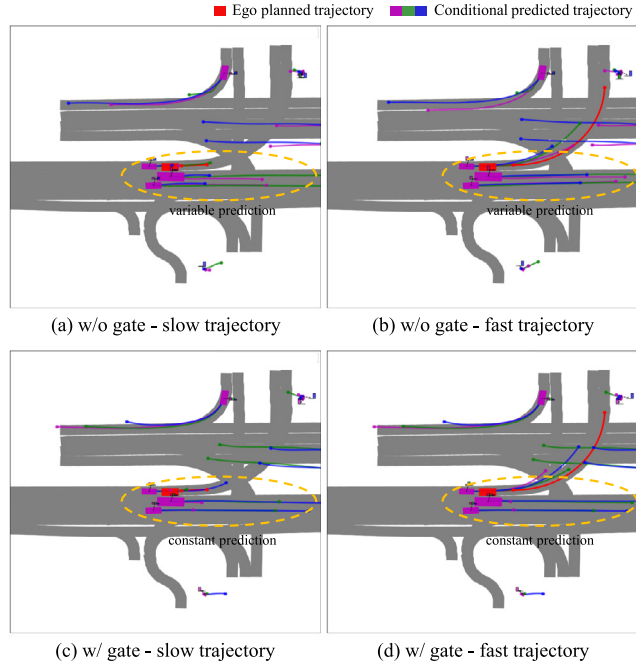


FIGURE 8. Qualitative results of conditioning gate network with a focus on the orange-circled regions. Without the gate network (a,b), ego conditioning unnecessarily affects the predicted trajectories of non-interacting agents. With the gate network (c,d), this effect is suppressed, allowing the model to focus ego conditioning on agents that are actually involved in interactions, resulting in more context-aware and reliable predictions.

effective at capturing ego-agent interactions, they suffer from substantial computational overhead and latency, limiting their applicability in real-time planning scenarios. To address this limitation, we incorporate multi-modal cross-attention, use a parallel conditioning architecture, and apply causal masked attention from DTPP [13], as it enables efficient single-pass inference conditioned on the entire set of ego plans. As shown in Table 3, our method reduces prediction latency by over 70% compared to its iterative counterpart, RaMP-iter.

3) EFFECTIVENESS OF CONDITIONING GATE NETWORK

We analyze the effect of the proposed conditioning gate network. Fig. 8 presents qualitative results comparing predictions without (top) and with (bottom) the conditioning gate network. Without the conditioning gate network (top), changes in the ego's future plan (from a slow to a fast trajectory) lead to spurious shifts in the predicted behaviors of surrounding agents—even those kinematically decoupled from the ego—resulting in non-causal and unreliable forecasts. In contrast, with the conditioning gate network, it enables reliable and semantically consistent predictions

TABLE 4. Quantitative results of conditioning gate network.

Method	Prediction accuracy	
	ADE (8s) ↓	FDE (8s) ↓
RaMP w/o gate	1.0594 m	2.4852 m
RaMP	0.9857 m	2.2789 m

by selectively injecting egocentric context only when it is behaviorally relevant. To complement this analysis, Table 4 reports quantitative results. The evaluation is based on the most probable predicted trajectory conditioned on the ego candidate closest to the ground truth. The conditioning gate achieves 7.5% and 9.0% improvements in Average Displacement Error (ADE) and Final Displacement Error (FDE), respectively, demonstrating its effectiveness in generating accurate, interaction-aware forecasts.

V. CONCLUSION

We propose a novel bidirectional RaMP framework that integrates a newly designed CMMP with TPP. RaMP effectively captures the reaction uncertainty of surrounding agents and probabilistically evaluates ego-trajectory candidates based on multi-modal prediction results. Qualitative results demonstrate RaMP's strong ability to capture reaction uncertainty. Closed-loop planning experiments, particularly under highly interactive conditions, further validate the effectiveness of our approach. We believe that RaMP offers a promising direction for reaction-aware motion planning, enabling safe, socially cooperative autonomy in highly interactive urban driving scenarios.

APPENDIX A

EVALUATION METRICS IN NUPLAN SIMULATION

This appendix describes the evaluation metrics used in the nuPlan benchmark [25] in detail. We first explain the overall scenario-level scoring procedure, followed by the definition and computation of each individual metric. For each scenario, the final evaluation score is computed as a combination of hard safety constraints and soft performance metrics. Specifically, the scenario score is defined as:

$$\text{Scenario Score} = \prod_{m \in \mathcal{M}_{\text{mult}}} m \times \frac{\sum_{i \in \mathcal{M}_{\text{score}}} w_i \cdot s_i}{\sum_{i \in \mathcal{M}_{\text{score}}} w_i} \quad (10)$$

where, $\mathcal{M}_{\text{mult}}$ denotes the set of multiplicative factors (hard constraints), $\mathcal{M}_{\text{score}}$ denotes the set of weighted metric scores, s_i is the score of metric i , and w_i is the corresponding weight. The multiplicative factors act as gating terms: if any critical safety or rule-violation metric fails, the entire scenario score is suppressed (i.e., becomes 0) regardless of performance on other metrics.

The remaining metrics are aggregated using a weighted average, capturing overall driving quality. The nuPlan evaluation framework separates metrics into multiplicative factors and weighted metric scores to jointly enforce safety and assess driving quality. Multiplicative factors correspond to

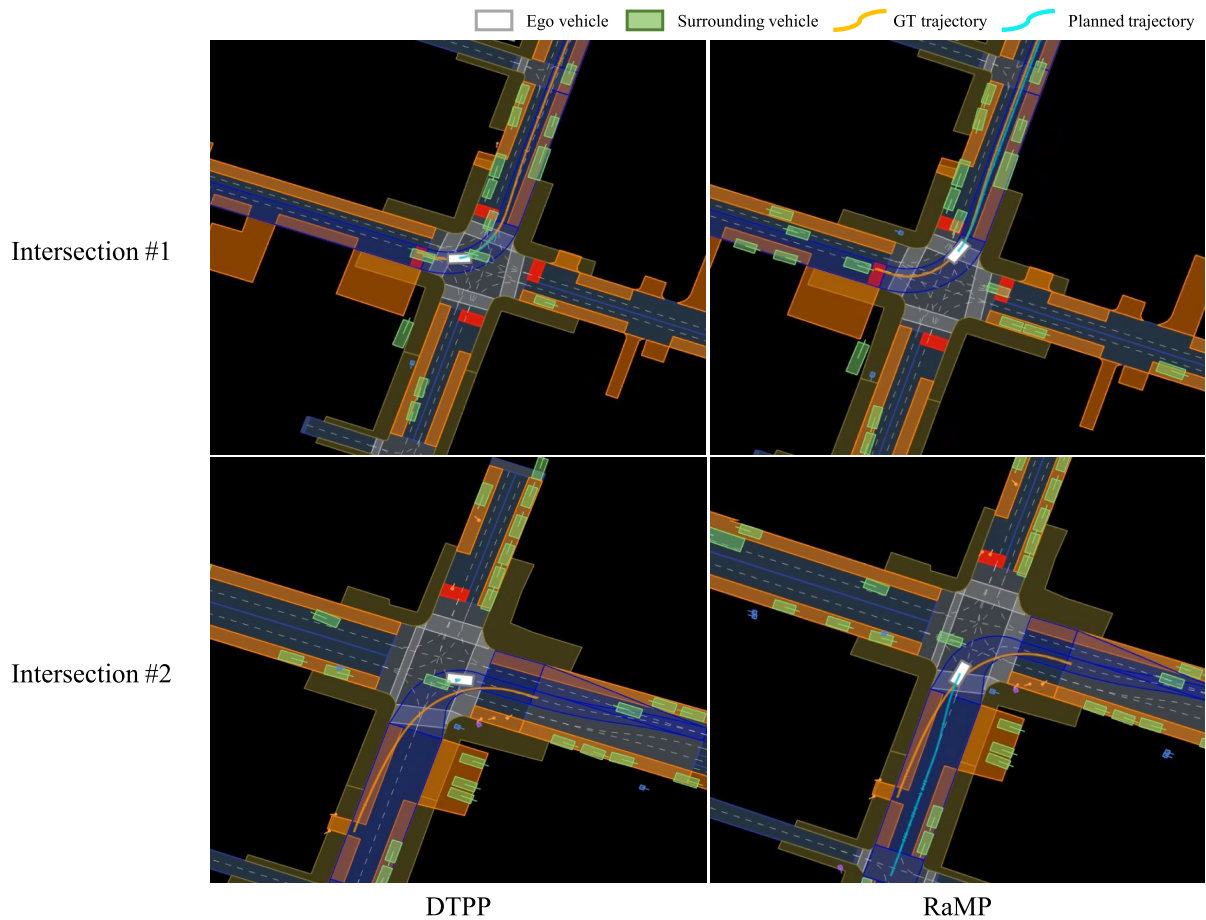


FIGURE 9. Qualitative comparison between the baseline and the proposed method on the nuPlan simulation in the intersection scenarios. In these intersection scenarios, the ego vehicle is required to make a left turn while considering interactions with surrounding vehicles. DTPP makes an incorrect decision due to overconfident assumptions about agent reactions, resulting in a collision. In contrast, by explicitly modeling reaction uncertainty, RaMP better captures interactive behaviors and makes a correct planning decision.

TABLE 5. nuPlan closed-loop metrics, scores and weights.

Metric name	Metric score
No at-fault Collisions	0, 0.5 or 1
Drivable Area Compliance	boolean
Driving Direction Compliance	0, 0.5 or 1
Making progress	boolean
Speed limit compliance	$f_{\text{speed-limit}}$
Time to Collision within bound	boolean
Progress along route	$\frac{\text{ego progress}}{\text{expert progress}}$
Comfort	boolean

hard constraints related to safety and rule compliance, such as at-fault collisions, drivable area compliance, driving direction compliance, and minimum progress. These metrics act as gating terms that suppress the entire scenario score when violated, regardless of performance on other criteria. In contrast, weighted metric scores capture continuous aspects of driving performance, including time-to-collision, progress along the expert's route, speed limit compliance, and comfort.

These metrics are aggregated using a weighted average to reflect overall driving quality once all hard constraints are satisfied. This separation ensures that unsafe behaviors are strictly penalized while still enabling fine-grained comparison among safe driving policies. Table 5 summarizes the nuPlan closed-loop evaluation metrics, including their score ranges and roles within the episode-level scoring framework.

APPENDIX B

QUALITATIVE PLANNING RESULTS ON THE SIMULATION

To demonstrate RaMP's ability to model interactions, we conducted nuPlan simulations in interactive scenarios. Fig. 9 and Fig. 10 illustrates the qualitative results in the intersection and crosswalk scenarios.

In Intersection #1, the ego vehicle executes an unprotected left turn from the left to the upward direction. In this scenario, another vehicle approaches from the upward direction, decelerating due to the ego vehicle, whereas another vehicle on the right remains stationary before moving. DTPP fails to account for the vehicle's deceleration in the upward direction. As a result, it slows down unnecessarily and loses the

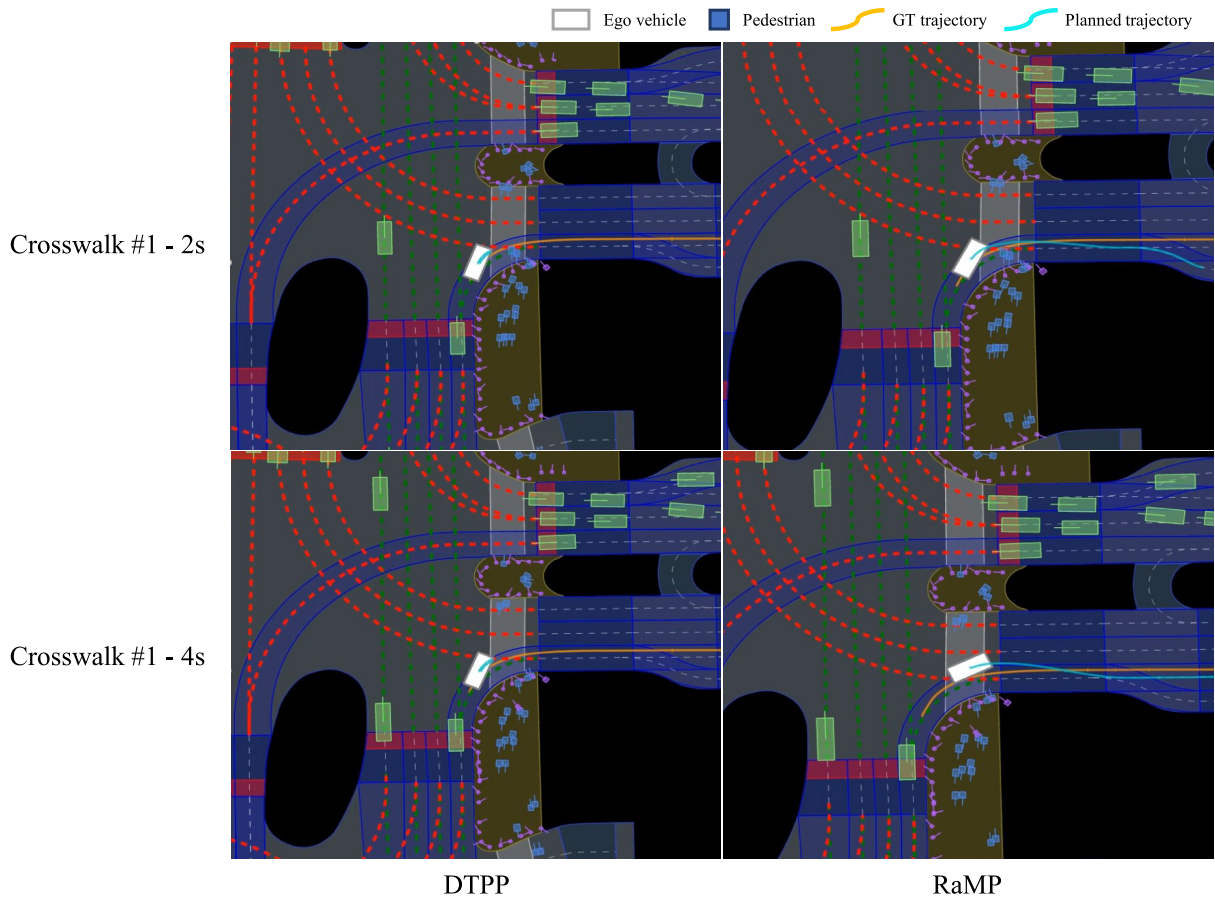


FIGURE 10. Qualitative comparison between DTPP and RaMP on the nuPlan simulation in the crosswalk scenario. In this scenario, the ego vehicle is required to execute a right turn while considering interactions with pedestrians crossing the crosswalk. DTPP waits until all pedestrians fully clear the crosswalk, whereas RaMP exploits both temporal and spatial gaps between pedestrians to proceed safely. As a result, the planned trajectory closely follows the expert trajectory while maintaining safety and efficiency.

appropriate time to cross the intersection. Additionally, as the right-side vehicle begins to move, the ego vehicle fails to fully stop to avoid a collision, instead moving slightly forward and ultimately colliding. In contrast, RaMP successfully anticipates the deceleration of the ego vehicle in the upward direction and proceeds through the intersection confidently. This behavior aligns with the expert trajectory (orange), demonstrating that the proposed method performs similarly to a human expert.

In Intersection #2, the ego vehicle performs an unprotected left turn from the right to the downward direction. In this scenario, an oncoming vehicle ignores the ego vehicle and continues driving without yielding. The baseline initially generates two crossing paths without slowing, likely assuming the oncoming vehicle would decelerate in response to the ego vehicle's actions. However, when it becomes apparent that the oncoming vehicle will not yield, the ego vehicle begins to decelerate but fails to stop in time, resulting in a collision. In contrast, RaMP predicts that the oncoming vehicle will proceed without yielding and confidently crosses the intersection without collision. Once

again, this behavior aligns with the expert trajectory, demonstrating the proposed method's ability to mimic expert-level decision-making.

In Crosswalk #1, the ego vehicle waits for pedestrians to cross. In this scenario, there are moments when gaps appear between pedestrians, allowing the vehicle to cross safely. However, the baseline waits for all pedestrians to completely clear the crosswalk before proceeding. In contrast, the proposed method understands both the timing and spatial gaps in pedestrian movement, enabling the ego vehicle to cross the crosswalk safely and efficiently during these intervals. This behavior closely matches the expert trajectory, further highlighting the method's human-like decision-making capabilities.

In summary, RaMP demonstrates superior comprehension of the effects of the ego vehicle's actions on surrounding agents compared to the baseline. The RaMP confidently navigates through intersections by understanding the impact of the ego plan on surrounding agents. These results confirm that RaMP effectively handles complex interactive scenarios by reasoning about the reactions of surrounding agents

and exhibiting expert-level understanding of how ego plans influence their behaviors.

ACKNOWLEDGMENT

The authors used ChatGPT (OpenAI) to improve the grammar and readability of the manuscript. The content and conclusion are entirely their own.

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